

Code No: 156BN**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year II Semester Examinations, August/September - 2021****MACHINE LEARNING****(Computer Science and Engineering)****Time: 3 Hours****Max. Marks: 75****Answer any five questions****All questions carry equal marks**

- 1.a) Define Well-Posed problem. Illustrate any four examples for Well-Posed problems.
b) What do you mean by Candidate elimination? Explain. [7+8]
- 2.a) What are the concepts of learning as search? Discuss.
b) Discuss the appropriate problems for decision tree learning. [8+7]
- 3.a) Contrast the hypothesis space search in ID3 and candidate elimination algorithm.
b) Explain the Back propagation learning algorithm and its limitations. [7+8]
- 4.a) How a multi layered network learns using a gradient descent algorithm? Discuss.
b) Explain the methods for comparing the accuracy of two hypotheses. [8+7]
- 5.a) State Bayes theorem. Illustrate Bayes theorem with an example.
b) Describe the mistake bound model of learning. [8+7]
- 6.a) Explain Gibbs algorithm with an example.
b) State and explain the Minimum Description Length Principle. [8+7]
- 7.a) Discuss about Hypothesis space search in genetic algorithms.
b) Write the basic algorithm for learning sets of First-Order Rules. [8+7]
- 8.a) Discuss Explanation-Based learning of search control knowledge.
b) Explain the inductive analytical approaches to learning. [8+7]

---ooOoo---

Code No: 156BN**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year II Semester Examinations, February - 2023****MACHINE LEARNING****(Computer Science and Engineering)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) Define Machine learning. [2]
- b) Write about inductive bias. [3]
- c) Define hypothesis. [2]
- d) Write about sampling theory. [3]
- e) Define Eager learning. [2]
- f) What is lazy Learning? [3]
- g) Define term Genetic. [2]
- h) Discuss Dynamic Programming. [3]
- i) Define explanation-based learning. [2]
- j) Write about control knowledge. [3]

PART – B**(50 Marks)**

2. Which disciplines have their influence on machine learning? Explain with examples. [10]

OR

- 3.a) Explain the two uses of features in machine learning.
- b) Discuss about decision tree representation, in detail. [5+5]

4. Explain in detail about Kernel Perceptron. [10]

OR

5. Discuss in detail about representation of Neural Networks. [10]

6. Describe briefly about k-nearest neighbor algorithm. [10]

OR

- 7.a) Discuss about Bayesian belief networks.
- b) Explain about Bayes theorem. [5+5]

8. Discuss Briefly about Genetic algorithms in detail. [10]

OR

- 9.a) Discuss about Q-learning, in detail.
- b) Explain temporal difference learning in detail. [5+5]

10. Explain about PROLOG-EBG, in detail. [10]

OR

11.a) Discuss about augment search operators.

b) Explain about search control knowledge in detail. [5+5]

---ooOoo---

Code No: 156BN**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year II Semester Examinations, February/March - 2022****MACHINE LEARNING****(Computer science and Engineering)****Time: 3 Hours****Max. Marks: 75****Answer any five questions****All questions carry equal marks**

- 1.a) Discuss about the basic Decision Tree Learning algorithm.
b) Briefly explain the need of Inductive Bias in decision Tree Learning. [7+8]
- 2.a) Explain the Find-S: Finding a Maximally Specific Hypothesis in detail.
b) Explain the issues in decision tree learning. [10+5]
- 3.a) Present the Backpropagation algorithm for feedforward networks and explain each step in it.
b) Explain how to estimate hypothesis accuracy. [9+6]
- 4.a) Define the terms estimation bias and confidence intervals.
b) Discuss the central limit theorem for deriving confidence intervals in detail.
c) Explain the representation of neural networks. [5+5+5]
- 5.a) Design the Brute Force Bayesian concept learning algorithm and elaborate.
b) Explain the Mistake Bound for the Halving Algorithm. [8+7]
- 6.a) Explain the Maximum Likelihood Hypotheses for predicting probabilities.
b) Elaborate the Locally Weighted Linear Regression. [8+7]
- 7.a) Explain the Q-learning with suitable example.
b) Explain about the hypothesis space search. [8+7]
- 8.a) Discuss about the Explanation-based Learning of Search Control Knowledge.
b) Explain how to initialize the hypothesis by using prior knowledge. [8+7]

---ooOoo---